8 A cylinder is made of modelling clay. The modelling clay is an electrical conductor.

Fig. 8.1 shows the cylinder.

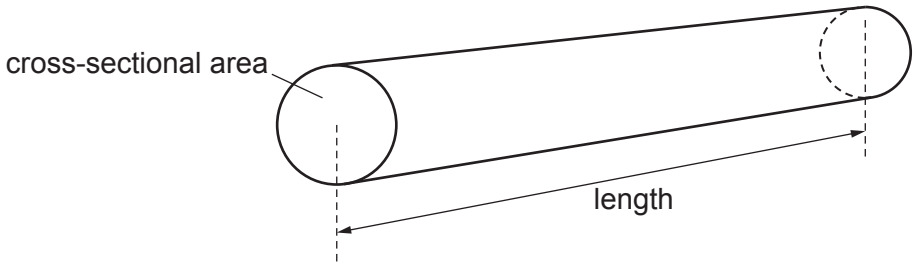


Fig. 8.1

The cylinder is connected into a circuit.

Fig. 8.2 shows that the circuit also includes a battery of electromotive force (e.m.f.) 9.0 V and a resistor P.

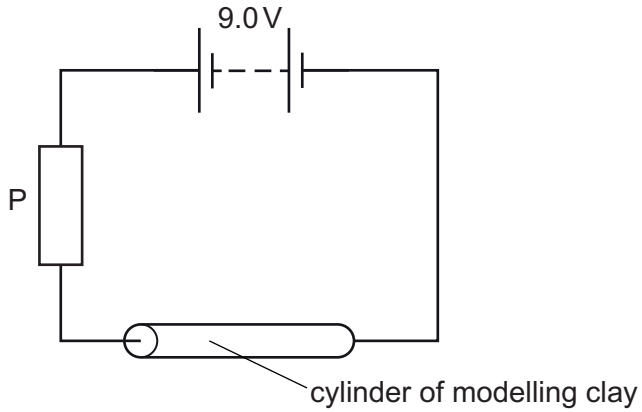


Fig. 8.2

The resistance of P is 4.0 Ω. The current in P is 1.5 A.

(a) Calculate:

(i) the magnitude *X* of the charge that flows through P in 600 s

X = ... [2]

(ii) the resistance of the cylinder of modelling clay.

resistance = ... [3]

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(b) The cylinder is removed from the circuit and replaced with a new cylinder made of the same modelling clay.

The new cylinder is twice the length and has half the cross-sectional area of the first cylinder.

Calculate the time that it now takes for a charge of magnitude *X* to flow through resistor P.

time = ... [4]

[Total: 9]